

GEOG 281A

Scale and the Openshaw Effect

Scale, it is more the MAUP

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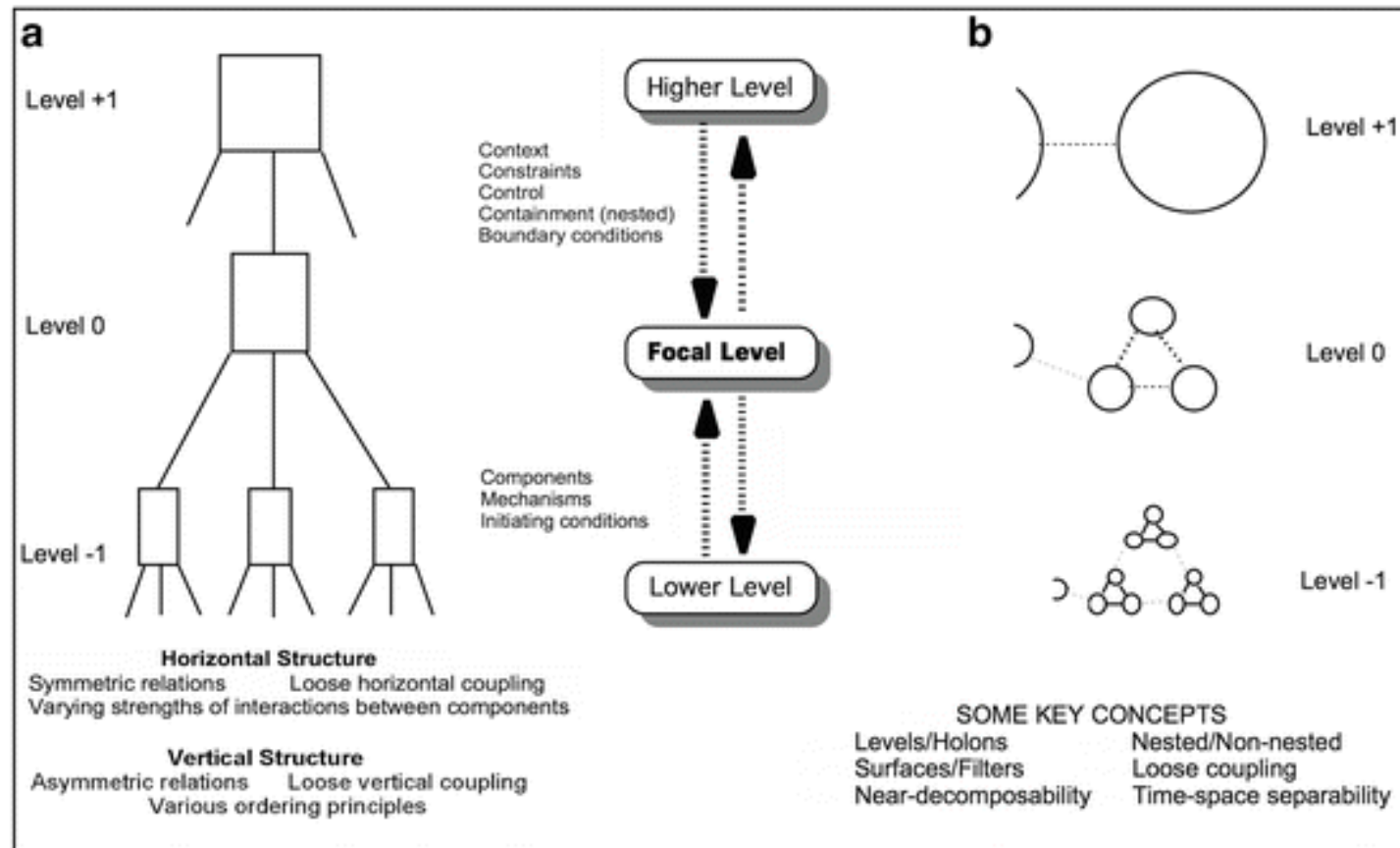
SCALE IN GEOGRAPHY & GISCIENCE

SCALE AS SIZE

Scale as the size of something, or as the extent of a study area

SCALE AS HIERARCHY

Scales as level is an interacting hierarchy defined by the processes and the decomposability of levels



STUDENT PERSPECTIVE

What would a research design look like if a researcher took non-decomposability seriously from the start?

SCALE AS A SOCIAL CONSTRUCTION

Scale as a product of human perspectives and actions that produce observable structure

STUDENT PERSPECTIVE

I would argue the opposite. Urban systems are deliberately planned and zoned through grid networks, land use designations, and administrative boundaries that create spatial hierarchies by design. We impose order on cities in ways that make them decomposable precisely because we built them to be.

If we consider decomposability is the measure of neatness, which systems are actually messier: the ones we engineered, or the ones we didn't?

SCALE AS STRUCTURED BY HUMAN COGNITION

Scale as a product of human perspectives and actions that produce observable structure

FIGURAL Spaces comprehended in a single view

VISTA Spaces from a single viewpoint, but larger than the body

ENVIRONMENTAL Spaces that must be understood through movement

GEOGRAPHIC Spaces that can only be understood using representation

STUDENT PERSPECTIVE

When does a space become too large to learn directly, thus requiring symbolic representation?

Is this classification system be better represented by a spectrum or continuum since these are subjective perspectives of space?

SCALE AS REPRESENTATIVE FRACTION

The ratio of the distance on a map to the distance in the world

The representative fraction breaks down when representing certain spatial extents, or when using certain projections

What happens with web mapping, or set-ups with multiple displays?

STUDENT PERSPECTIVE

As geographers and spatial scientists, do we seek to make discoveries that are invariant to scale? Or, is scale-dependent knowledge also valuable?

BREAK

THE OPENSHAW EFFECT

OPENSHAW'S FIRST EXPERIMENT

SCALE

Variation in results observed when the same data are combined into sets of increasingly larger units

AGGREGATION

Variation in results where the number of units n is held constant

OPENSHAW'S FIRST EXPERIMENT

Alternative combinations of counties	<i>r</i>
6 Republican-proposed congressional districts	0.4823
6 Democrat-proposed congressional districts	0.6274
6 Congressional districts	0.2651
6 Urban/rural regional types	0.8624
6 Functional regions	0.7128
99 Iowa counties	0.3466

ACTIVITY

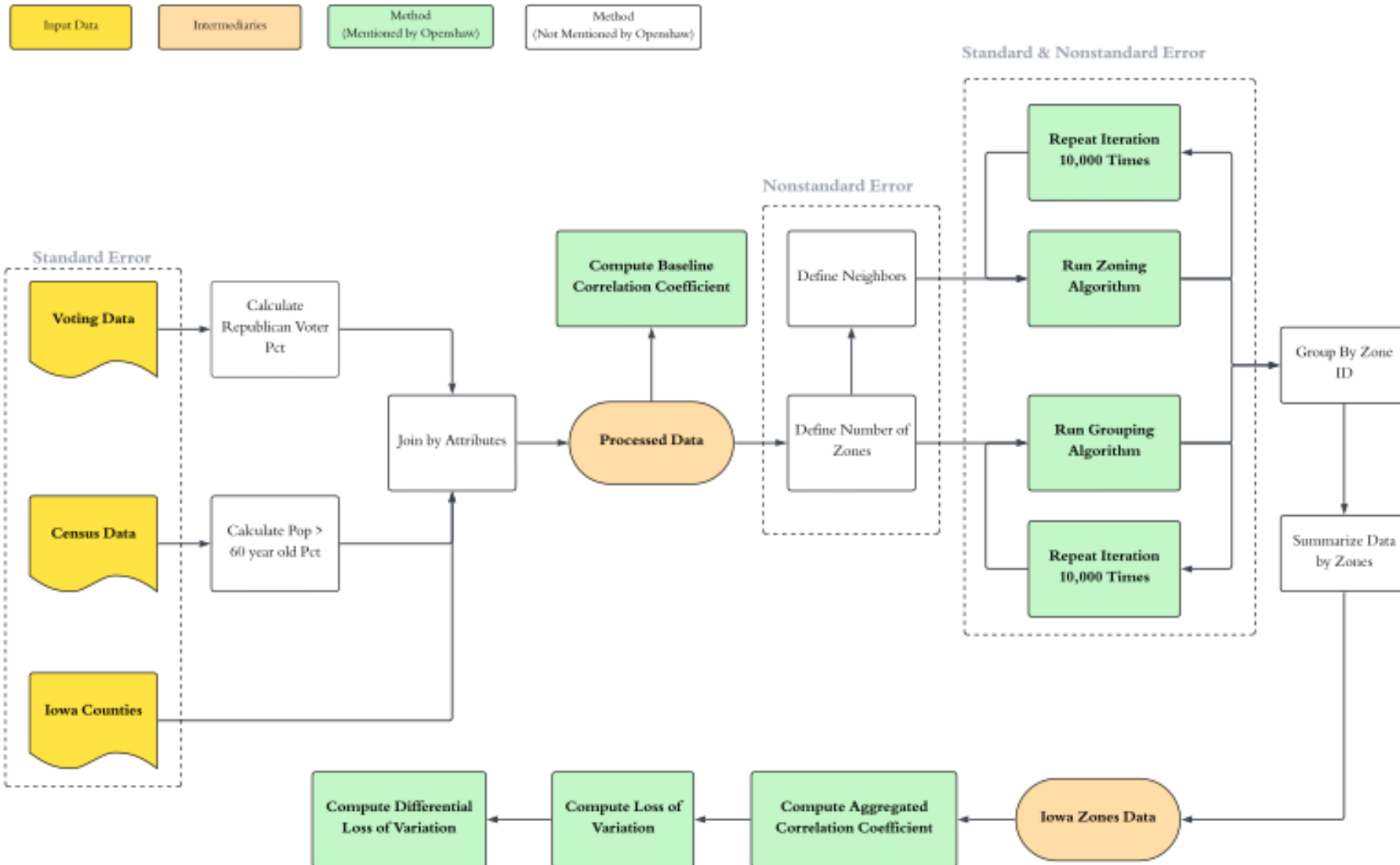
Purpose: To assess your understanding of the original Openshaw experiment and practice research design and planning

Task:

- In groups of three, assemble the provided workflow components into a diagram that reflects the sequence of steps in Openshaw's experiment.
- Draw connections between components and add any missing steps you believe are necessary.

DEBRIEF

- Compare your workflow with the workflows of other groups.
- Which steps in the workflow are included in Openshaw, which are left for the reader to interpret?
- Where is uncertainty present in the experiment? What kinds of error is it creating?
- How is scale being defined and manipulated in this experiment?
- If you could redesign this experiment, what would you do differently?

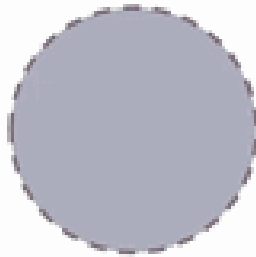


A SIMPLE TAXONOMY OF ERRORS

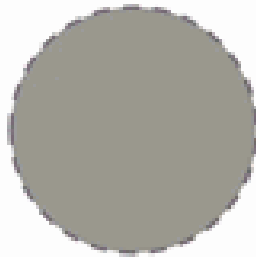
STANDARD	Variation that arises from natural variation in phenomena and sampling
NONSTANDARD	Variation that arises because there is generally no 'standard' path through an analysis, which requires researchers to make decisions

DATASET

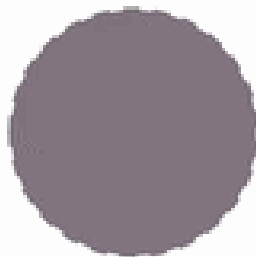
Sample 1



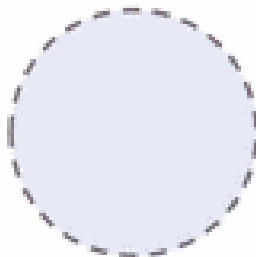
Sample 2



Sample 3



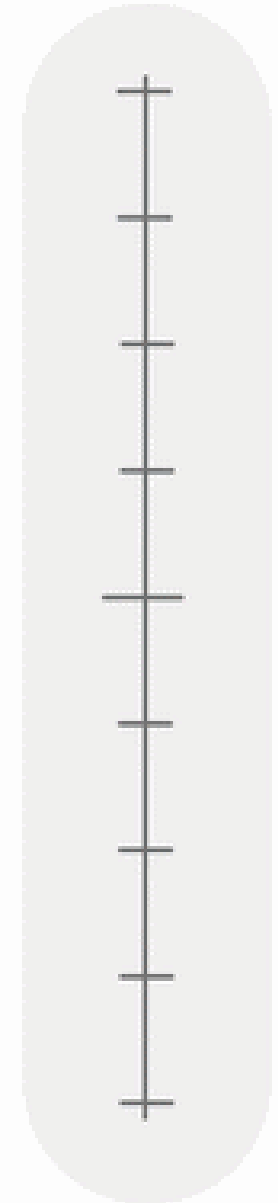
Sample 4



ANALYSIS



RESULTS



ANALYTICAL DECISIONS

RESULTS



BREAK

ACTIVITY

Purpose: Directly observe the Openshaw Effect and generate a distribution of the magnitude of that effect

SETUP

Create a Local Copy of the Repository:

- Please work in pairs. At least one person in each pair should have the following already set up on their local machine: a GitHub account, Git/GitHub installed, and RStudio installed.
- Go to the course webpage. In the left-side menu bar, navigate to the Scale lesson, then open the Openshaw Effect lesson. In that lesson, you will find two links that we will use for today's activity: the GitHub repository that contains the activity materials, and the Google Sheet where we will record and compare our correlation coefficients.
- Open the GitHub repository linked from the lesson page. Click the Fork button in the upper-right corner of the GitHub page. This will create your own copy of the repository under your personal GitHub account. Make sure you are forking the repository named Learn-MAUP.
- After you have forked the repository, clone your forked copy to your local machine. To do this, go to your forked repository on GitHub, click the green Code button, copy the repository URL, and use GitHub Desktop, the terminal, or another Git tool to clone it to a folder on your computer. Make sure you remember where you saved the cloned repository.

SETUP

Open and Run the Code:

- Once the repository has been cloned, navigate to the folder on your computer where you saved it. Inside the folder, find the R Project file. It should end in .Rproj. Open this file , not the .Rmd file or any other file first. Opening the .Rproj file will launch RStudio and automatically set the correct working directory to the folder containing the activity materials.
- After RStudio opens, use the Files pane to go to the data folder and open In-Class Activity.Rmd. This is the file we will work through during class.
- Start running the code from top to bottom. Read the instructions in the R Markdown file carefully as you go. When the file asks you to modify something, record a value, follow the instructions before moving on. At the end of the activity, enter your group's correlation coefficient into the shared Google Sheet so that we can compare results across the class.

DEBRIEF

- Interpret the results of the experiment (scale problem, aggregation problem, n)
- GIScience foundations and resulting decisions

BREAK

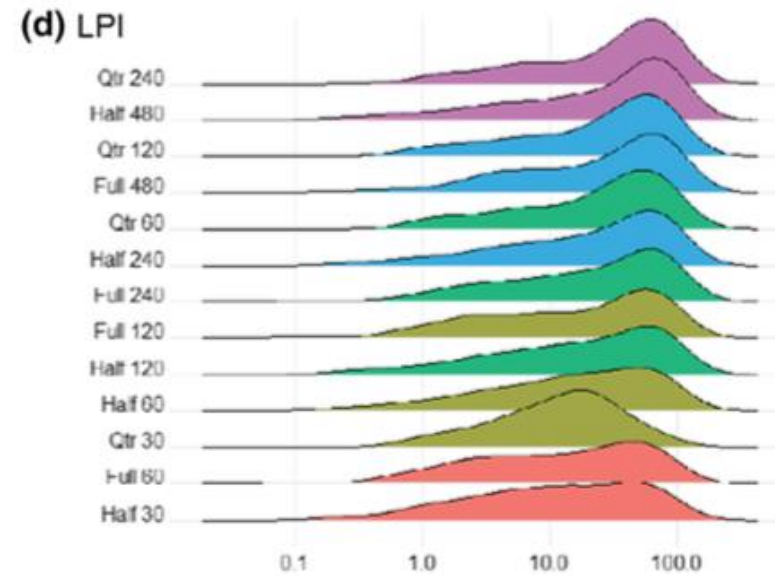
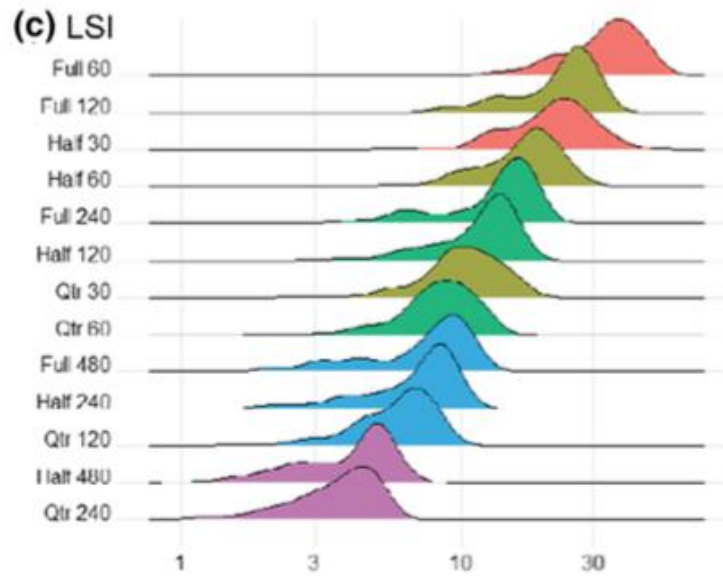
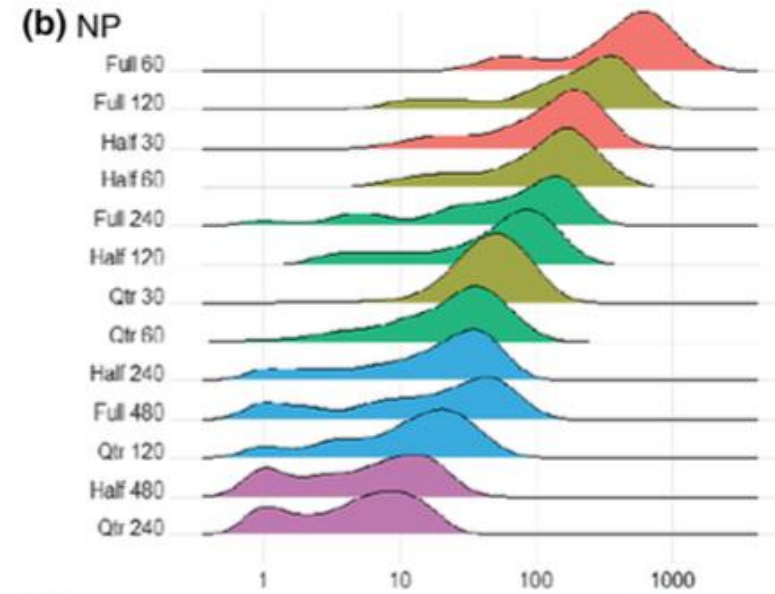
SCOPE

SCOPE

The dimensionless ratio between the range (extent) of a representation and the resolution (grain) of that representation

(a)

	Extent		
Grain km ² (res. m)	25 km ² (Qtr)	100 km ² (Half)	400 km ² (Full)
0.0009 (30 m)	B	A	
0.0036 (60 m)	C	B	A
0.0144 (120 m)	D	C	B
0.0576 (240 m)	E	D	C
0.2304 (480 m)		E	D



PRACTICAL ADVICE FROM DR FRAZIER

- FIRST** Researchers should set scope based on the phenomenon being studied, rather than relying solely on available data.
- SECOND** Studies should report both grain and extent quantitatively so that scope can be calculated and compared across studies.
- THIRD** Researchers should undertake comparisons and replications based on scope.

STUDENT PERSPECTIVE

Should spatial objects have the ability to preserve native resolutions after steps like aggregation? What implications would this have for replicability in research?